

ORIGINAL CONTRIBUTION

Ischemic Postconditioning in Patients With Acute Stroke With Successful Recanalization After Endovascular Therapy: A Pilot Randomized Controlled Trial

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BACKGROUND: Ischemia-reperfusion injury contributes to continued infarct growth despite successful recanalization. Ischemic postconditioning (IPostC) reduces infarct size in preclinical models, but its therapeutic effect as an adjunct to endovascular thrombectomy remains uncertain. We investigated whether IPostC performed after successful recanalization reduces infarct growth in patients with acute ischemic stroke.

METHODS: This single-center, prospective, randomized, open-label trial with blinded end point assessment was conducted at Tianjin Huanhu Hospital between September 2024 and May 2025. Patients with acute ischemic stroke who underwent endovascular thrombectomy within 24 hours of symptom onset or last known well, had a baseline National Institutes of Health Stroke Scale score ≥ 6 and a prestroke modified Rankin Scale score ≤ 2 , and achieved successful recanalization after endovascular thrombectomy were randomly assigned to the IPostC or control group. IPostC consisted of 4 cycles of 2-minute balloon inflation followed by 2-minute deflation, with the balloon positioned at the original intracranial arterial occlusion site. The primary outcome was infarct growth from baseline to 48-hour magnetic resonance imaging (MRI). Secondary outcomes included infarct volume on immediate postprocedure MRI (within 2 hours) and on 48-hour MRI; early infarct growth and late infarct growth; serial National Institutes of Health Stroke Scale scores during hospitalization; and modified Rankin Scale score at 90 days. Primary and secondary outcomes were analyzed in the intention-to-treat population using unadjusted between-group comparisons.

RESULTS: Sixty patients were enrolled, with 30 randomly assigned to each group. Infarct growth from baseline to 48-hour MRI was lower with IPostC than with control (median difference, -8.7 mL [95% CI, -17.1 to -1.4]; $P=0.015$). In the serial MRI subgroup ($n=30$; 15 per group), early infarct growth was also lower with IPostC (median difference, -3.2 mL [95% CI, -8.7 to -0.1]; $P=0.045$). Patients in the IPostC group had a lower National Institutes of Health Stroke Scale score at 24 hours (median difference, -3.0 [95% CI, -5.0 to -1.0]; $P=0.005$). The 90-day modified Rankin Scale distribution did not differ significantly between groups (common odds ratio, 1.1 [95% CI, 0.4–2.6]; $P=0.921$).

CONCLUSIONS: In this pilot randomized trial, IPostC performed after successful recanalization reduced infarct growth in patients with acute ischemic stroke due to large-vessel occlusion. These findings require confirmation in larger randomized trials.

REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT06545734.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: cytoprotection ■ infarction ■ ischemic stroke ■ reperfusion injury ■ thrombectomy

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Supplemental Material is available at <https://www.ahajournals.org/doi/suppl/10.1161/STROKEAHA.125.056167>.

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Stroke is available at www.ahajournals.org/journal/str

Nonstandard Abbreviations and Acronyms

BBB	blood-brain barrier
DCE-MRI	dynamic contrast-enhanced magnetic resonance imaging
DWI	diffusion-weighted imaging
EVT	endovascular thrombectomy
IPostC	ischemic postconditioning
MRI	magnetic resonance imaging
mRS	modified Rankin Scale
NIHSS	National Institutes of Health Stroke Scale

Endovascular thrombectomy (EVT) reduces infarct growth and improves clinical outcomes.¹ However, despite high rates of successful recanalization, only approximately half of treated patients achieve functional independence.^{2,3} Continued infarct growth after recanalization,^{4–6} driven in part by ischemia-reperfusion injury, may contribute to incomplete functional recovery. Cytoprotective strategies aimed at limiting continued infarct growth may therefore represent a promising adjunct to EVT.

Ischemic postconditioning (IPostC), performed at the onset of reperfusion, has been shown to reduce infarct volume and improve neurological outcomes in preclinical stroke models,^{7–10} with proposed mechanisms including attenuation of oxidative stress, inflammation, apoptosis, and blood-brain barrier (BBB) disruption.^{11,12} However, clinical implementation of IPostC has been constrained by technical and device limitations.

EVT restores blood flow and creates the procedural conditions needed for immediate IPostC after successful recanalization¹³; the procedure consists of brief balloon inflation-deflation cycles, with inflation temporarily interrupting antegrade flow and deflation restoring reperfusion. However, protocols have varied across early clinical studies.^{14–16} One externally controlled exploratory study used 4 cycles of 2-minute in situ balloon inflation-deflation and suggested less deep infarct growth.¹⁵ Another ambidirectional cohort study used 5 cycles of 15-second balloon inflation-deflation in the ipsilateral internal carotid artery (ICA) and reported higher 90-day functional independence (54% versus 34%; post hoc number needed to treat of 5), along with reduced infarct growth and less cerebral edema.¹⁶ These nonrandomized clinical studies represent important early progress, but randomized evidence remains limited.

Against this background, we conducted a pilot randomized trial to test whether IPostC performed immediately after successful recanalization reduces infarct growth and improves clinical outcomes.

METHODS

The data that support the findings of this study are available from the corresponding author on reasonable request.

Study Design and Participants

We conducted a single-center, prospective, randomized, open-label trial with blinded end point assessment at Tianjin Huanhu Hospital. This report follows the CONSORT 2025 reporting guideline for randomized trials.¹⁷ The completed CONSORT checklist and the study protocol are provided in the [Supplemental Material](#). The study protocol was approved by the institutional review board (protocol no. 2024-213). Written informed consent was obtained from all patients or their legally authorized representatives.

Eligible patients were adults aged ≥ 18 years with acute ischemic stroke who underwent EVT within 24 hours after symptom onset or last known well, had a baseline National Institutes of Health Stroke Scale (NIHSS) score of 6 or higher, and were functionally independent before stroke, defined as a prestroke modified Rankin Scale (mRS) score of 0 to 2. All patients underwent baseline magnetic resonance imaging (MRI), had an Alberta Stroke Program Early Computed Tomography Score of 6 to 10, and had unilateral M1 or intracranial ICA occlusion. Successful recanalization after thrombectomy was confirmed by digital subtraction angiography and defined as an expanded Thrombolysis in Cerebral Infarction grade of 2b to 3. Patients who required stent implantation or >2 angioplasty attempts to achieve successful recanalization were excluded. A complete list of eligibility criteria is provided in the [Supplemental Methods](#).

Randomization and Blinding

After angiographic confirmation of successful recanalization after EVT, eligible patients were randomly assigned in a 1:1 ratio to the IPostC or control group. Patients assigned to the IPostC group immediately underwent in situ IPostC. For patients assigned to the control group, no IPostC was performed, and the endovascular procedure was then concluded. The random allocation sequence was computer-generated and implemented through an interactive Web response system, with stratification by age and baseline NIHSS score.

Because of the procedural nature of IPostC, treating neurointerventionists were not blinded to treatment assignment. Clinical outcomes were assessed by evaluators who were not involved in the endovascular procedure and were blinded to treatment allocation. These evaluators were not provided with randomization records or procedural information indicating whether IPostC had been performed before outcome assessment. Imaging outcomes were assessed by blinded evaluators using deidentified imaging data sets labeled only by study number and imaging time point.

IPostC Procedure

Reperfusion was achieved using standard thrombectomy techniques, including stent-retriever thrombectomy, aspiration thrombectomy, or a combination of both, at the discretion of the neurointerventionist. After successful recanalization, IPostC

was performed using the Neuro LPS low-pressure balloon catheter (SINOMED). To minimize vascular injury, the shortest balloon length and the smallest appropriate diameter were selected. The balloon was positioned in situ at the originally occluded arterial segment, as determined by prethrombectomy angiography.

The IPostC protocol consisted of 4 cycles of balloon inflation and deflation. In each cycle, the balloon was inflated at the minimum pressure required to arrest antegrade flow for 2 minutes, followed by 2 minutes of balloon deflation to restore reperfusion (Figure 1A). A prior dose-escalation study established the safety and tolerability of this inflation-deflation cycle.¹⁴ After completion of IPostC, angiography in anteroposterior and lateral projections was performed to assess final reperfusion status. If the expanded Thrombolysis in Cerebral Infarction grade was unchanged or improved compared with the pre-IPostC assessment, no further intervention was performed. If antegrade flow deteriorated, rescue therapy was initiated at the neurointerventionist's discretion and could include

balloon angioplasty, stent implantation, intra-arterial thrombolysis, or repeat thrombectomy.

Follow-Up

Patients were followed for 3 months. Both groups received standard care according to institutional protocols. Vital signs, physical examination findings, neurological assessments, and adverse events were recorded during follow-up. Immediate postprocedure MRI, including diffusion-weighted imaging (DWI), was performed within 2 hours after the procedure. Follow-up MRI was performed at 48 hours and included DWI, fluid-attenuated inversion recovery, dynamic contrast-enhanced MRI (DCE-MRI), and 3-dimensional time-of-flight MR angiography. These sequences were used to assess infarct volume, BBB permeability, and target-vessel recanalization status. The mRS score was assessed at 3 months, preferably in person. When in-person assessment was not feasible, the mRS score was determined by video or telephone interview with

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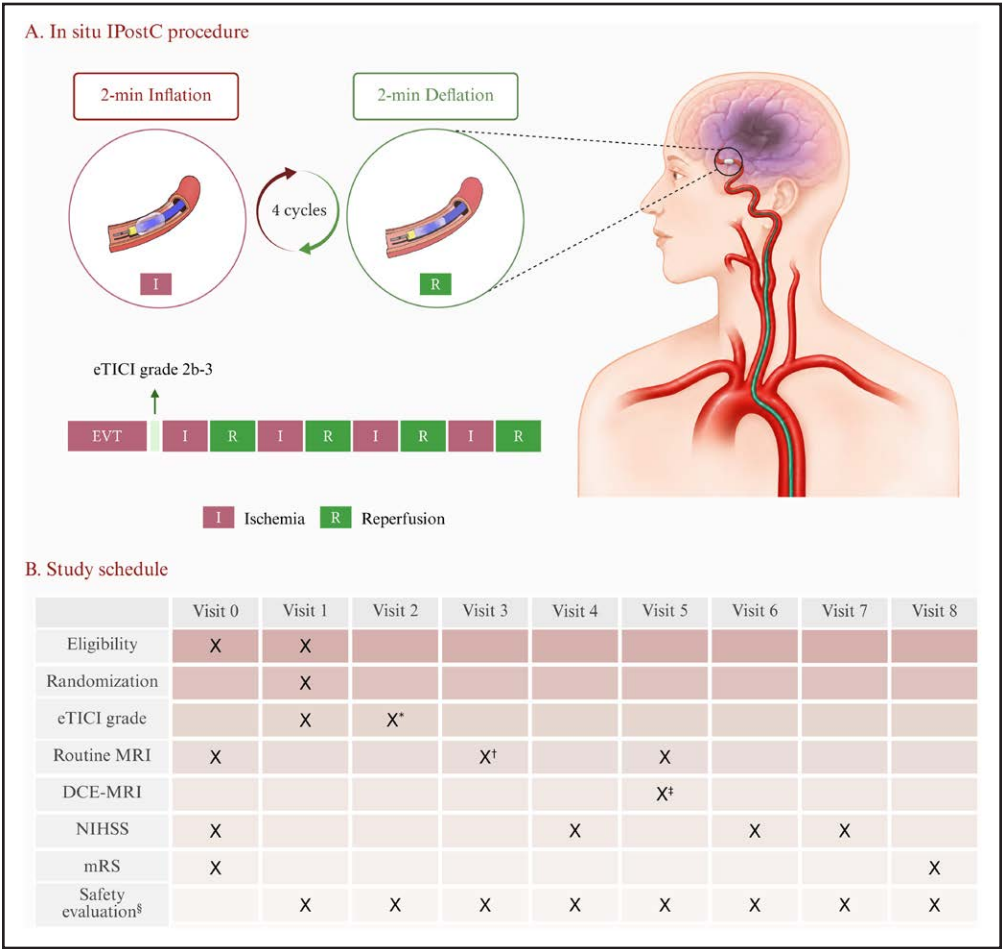


Figure 1. Overall study design. **A**, Schematic of the in situ ischemic postconditioning (IPostC) procedure. **B**, Study schedule. Visit 0 indicates baseline; visit 1, after endovascular thrombectomy (EVT); visit 2, during the IPostC procedure; visit 3, within 2 hours after the procedure; visit 4, day 1; visit 5, day 2; visit 6, day 3; visit 7, days 5 to 7 or at discharge; and visit 8, day 90. eTICI indicates expanded Thrombolysis in Cerebral Infarction; mRS, modified Rankin Scale; and NIHSS, National Institutes of Health Stroke Scale. *The visit 2 assessment refers to procedural angiographic assessment. †The visit 3 routine magnetic resonance imaging (MRI) was performed in the serial MRI subgroup when clinically feasible and did not include magnetic resonance angiography. ‡Dynamic contrast-enhanced magnetic resonance imaging (DCE-MRI) was performed at visit 5 in the serial MRI subgroup. §Safety was monitored from randomization through 90 days.

the patient, a relative, or a healthcare provider. The follow-up schedule is shown in Figure 1B.

Study Outcomes

The primary outcome was infarct growth from baseline to 48 hours, defined as the difference in infarct volume between baseline MRI and 48-hour MRI. Baseline infarct volume was measured on DWI, and 48-hour infarct volume was measured on fluid-attenuated inversion recovery.

Secondary imaging outcomes included infarct volume on immediate postprocedure MRI and on 48-hour MRI. Immediate postprocedure infarct volume was measured on DWI performed within 2 hours after the procedure. Early and late infarct growth were also assessed as secondary outcomes in the serial MRI subgroup, which included patients who underwent MRI at baseline, within 2 hours after the procedure, and at 48 hours. Early infarct growth was defined as growth from baseline to the immediate postprocedure MRI, and late infarct growth as growth from the immediate postprocedure MRI to 48-hour MRI.⁵

Infarct volume was quantified using 3-dimensional Slicer (www.slicer.org), as previously described.¹⁸ The infarct region was segmented on the relevant MRI sequence at each time point and manually refined slice by slice to exclude artifacts, hemorrhage, leukoaraiosis, and chronic infarcts. Final 3-dimensional infarct masks were used to calculate infarct volume.

Secondary clinical outcomes included the 90-day mRS score distribution; good outcome, defined as an mRS score 0 to 2; excellent outcome, defined as an mRS score of 0 to 1; early neurological improvement, defined as a reduction of ≥ 8 points in the NIHSS score within 24 hours; and serial NIHSS scores at 24 hours, day 3, and days 5 to 7 after randomization or at discharge.

IPostC-related procedural safety outcomes included arterial dissection, vessel perforation, distal embolization, and vasospasm. Other safety outcomes included final postprocedure reperfusion status assessed by expanded Thrombolysis in Cerebral Infarction grade, sustained target-vessel recanalization on 48-hour magnetic resonance angiography, all-cause mortality at 90 days, symptomatic intracranial hemorrhage, and any intracranial hemorrhage.

DCE-MRI Protocol

Because BBB disruption may contribute to continued infarct growth after EVT,¹⁹ we assessed BBB permeability using DCE-MRI performed at 48 hours to quantify BBB integrity. All patients were scanned on a 3.0-T system (MAGNETOM Prisma, Siemens Healthineers, Erlangen, Germany) with a 64-channel head coil. DCE-MRI postprocessing was performed using a standard Tofts model-based approach, with K_{trans} used as the permeability metric.

Complete MRI acquisition parameters and details of motion correction, arterial input function selection, regional-of-interest placement, and K_{trans} quantification are provided in [Supplemental Methods](#); representative K_{trans} maps are shown in [Figure S1](#).

Statistical Methods

All efficacy analyses were performed in the intention-to-treat population, and safety analyses were performed in the

as-treated population. Analyses were performed using available data, without imputation for missing values. Statistical analyses were performed using SPSS version 26.0 (IBM Corp, Armonk, NY) and R version 4.3.2. All statistical tests were 2-sided, and a $P < 0.05$ was considered statistically significant.

Given the exploratory nature of this pilot randomized trial, all prespecified efficacy analyses were unadjusted and considered hypothesis-generating. No adjustments for multiple comparisons were applied; therefore, findings should be interpreted with caution. No interim analyses were planned. Because this was a pilot study, no formal sample size or power calculation was performed. The target sample size of 60 patients was based on the anticipated recruitment capacity at the study center.

Continuous variables are reported as mean (SD) or median (interquartile range [IQR]), as appropriate, and categorical variables as number (percentage). Normality was assessed using the Shapiro-Wilk test and visual inspection of distributional diagnostics. Between-group comparisons of continuous outcomes were performed using *t* test when assumptions of approximate normality and homoscedasticity were met; otherwise, the Mann-Whitney *U* test was used, with the Hodges-Lehmann estimator used to estimate median differences and corresponding 95% CIs. Categorical outcomes were compared using the χ^2 test or Fisher exact test, as appropriate, with risk ratios and 95% CIs derived from contingency tables.

All adjusted analyses were nonprespecified post hoc analyses. Post hoc adjusted analyses of infarct growth were performed using median regression models adjusted for baseline infarct volume and onset-to-recanalization time. Post hoc longitudinal analyses of follow-up NIHSS scores were performed using linear mixed-effects models with treatment group, visit, and baseline NIHSS score as fixed effects and a patient-level random intercept.

RESULTS

Baseline Characteristics

Between September 2024 and May 2025, 128 patients were screened at Tianjin Huanhu Hospital. Of these, 60 patients were enrolled and randomized, with 30 assigned to the IPostC group and 30 to the control group. The CONSORT flow diagram is provided in [Figure 2](#). All participants received the assigned intervention, and no crossovers occurred. Baseline and 48-hour MRI data required for the primary outcome were available for all 60 randomized patients. Immediate postprocedure MRI outcomes were analyzed in the serial MRI subgroup ($n=30$; 15 per group), as specified in the protocol. 90-day mRS score data were available for all 60 randomized patients.

Baseline demographic and clinical characteristics are summarized in [Table 1](#) and were generally balanced between the 2 groups. The mean age was 63.3 ± 9.6 years, and 11 participants (18.3%) were female. The median admission NIHSS score was 12 (IQR, 10.8–14), and the median baseline infarct volume was 9.4 mL (IQR, 5.2–17.0). Overall, 50 patients (83.3%) had an M1 occlusion,

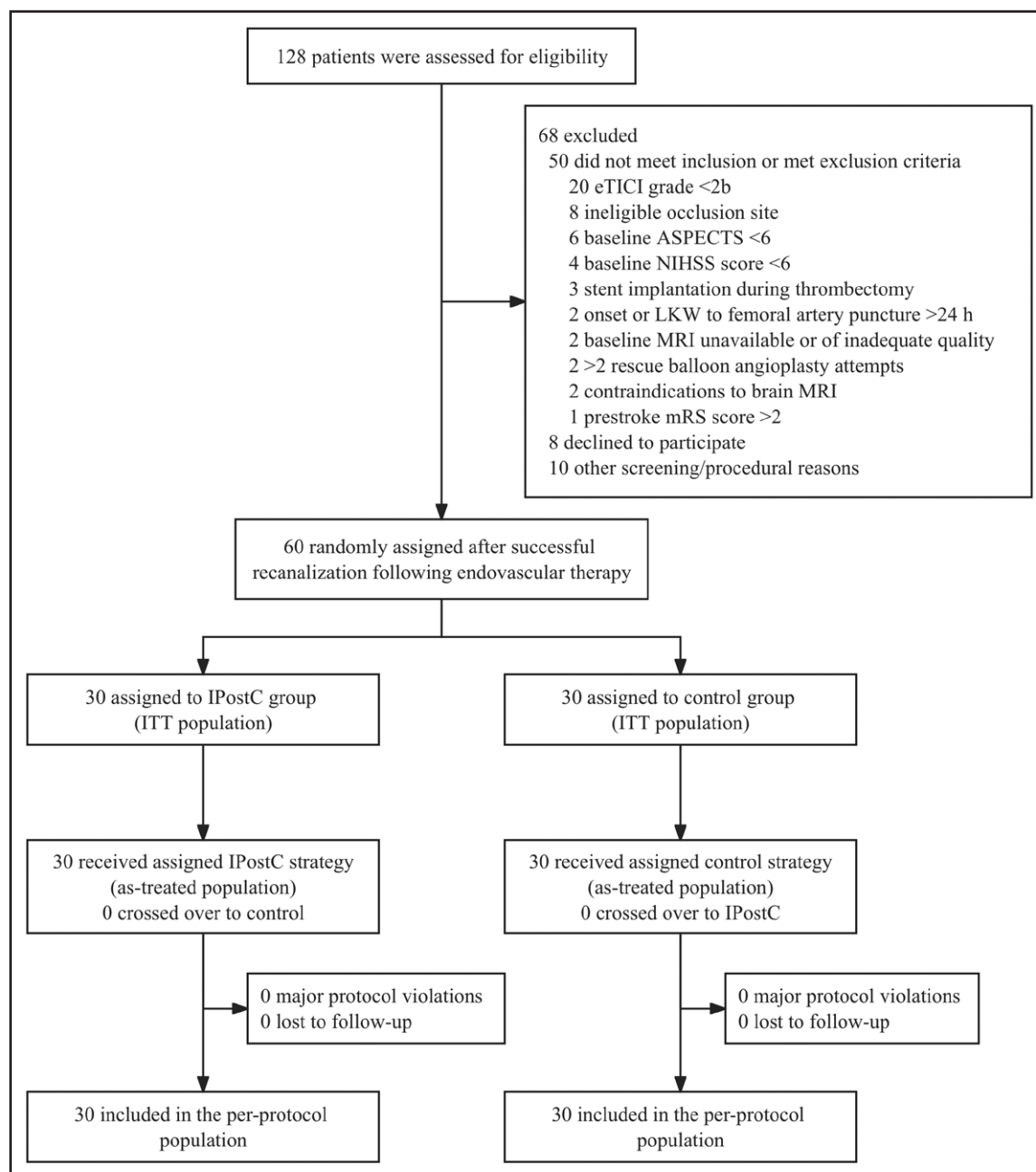


Figure 2. Trial profile.

ASPECTS indicates Alberta Stroke Program Early Computed Tomography Score; eTICI, expanded Thrombolysis in Cerebral Infarction; IPostC, ischemic postconditioning; ITT, intention-to-treat; LKW, last known well; MRI, magnetic resonance imaging; mRS, modified Rankin Scale; and NIHSS, National Institutes of Health Stroke Scale.

7 (11.7%) had tandem lesions, and 3 (5.0%) had intracranial ICA occlusion. Large-artery atherosclerosis was the presumed stroke cause in 48 patients (80.0%). 3 patients (5.0%) had a prestroke mRS score of 1.

Primary Outcome

Infarct growth from baseline to 48 hours was significantly lower in the IPostC group than in the control group (median [IQR], 10.5 mL [3.6–16.8] versus 20.1 mL

[9.7–32.3]; median difference, −8.7 mL [95% CI, −17.1 to −1.4]; $P=0.015$; Figure 3A). Patient-level infarct volume trajectories from baseline to 48 hours are shown for the control and IPostC groups in Figure 3B and 3C, respectively. In a nonprespecified post hoc median regression model adjusted for baseline infarct volume and onset-to-recanalization time, the treatment effect remained consistent with the primary unadjusted analysis (adjusted median difference, −9.2 mL [95% CI, −18.2 to −0.1]; $P=0.047$).

Table 1. Baseline Characteristics

	Total	IPostC	Control
No. of patients	60	30	30
Demographics			
Age, mean (SD), y	63.3 (9.6)	64.4 (8.4)	62.1 (10.7)
Female, n (%)	11 (18.3)	7 (23.3)	4 (13.3)
Clinical and imaging characteristics			
Admission NIHSS score, median (IQR)	12 (10.8–14)	12 (9.3–14)	12.5 (11–14)
Prestroke mRS score, n (%)			
0	57 (95.0)	27 (90.0)	30 (100.0)
1	3 (5.0)	3 (10.0)	0 (0.0)
ASPECTS, median (IQR)	8 (7–9)	8 (7–9)	8 (7–8.8)
Baseline infarct volume, median (IQR), mL	9.4 (5.2–17.0)	10.3 (5.8–18.7)	8.2 (4.7–16.4)
Occlusion site, n (%)			
M1 segment	50 (83.3)	24 (80.0)	26 (86.7)
Intracranial ICA	3 (5.0)	1 (3.3)	2 (6.7)
Tandem lesion	7 (11.7)	5 (16.7)	2 (6.7)
TOAST classification, n (%)			
CE	12 (20.0)	5 (16.7)	7 (23.3)
LAA	48 (80.0)	25 (83.3)	23 (76.7)
Medical history			
Hypertension, n (%)	35 (58.3)	17 (56.7)	18 (60.0)
Diabetes, n (%)	13 (21.7)	9 (30.0)	4 (13.3)
Previous stroke, n (%)	10 (16.7)	6 (20.0)	4 (13.3)
CAD, n (%)	11 (18.3)	6 (20.0)	5 (16.7)
AF, n (%)	14 (23.3)	9 (30.0)	5 (16.7)
Heart failure, n (%)	0 (0.0)	0 (0.0)	0 (0.0)
IV thrombolysis, n (%)	8 (13.3)	4 (13.3)	4 (13.3)
Antiplatelet use, n (%)	17 (28.3)	10 (33.3)	7 (23.3)
Anticoagulant use, n (%)	6 (10.0)	3 (10.0)	3 (10.0)
Current smoking, n (%)	38 (63.3)	19 (63.3)	19 (63.3)
Alcohol use, n (%)	32 (53.3)	18 (60.0)	14 (46.7)
Admission status			
Transfer status, n (%)			
Direct admission	55 (91.7)	29 (96.7)	26 (86.7)
Interhospital transfer	5 (8.3)	1 (3.3)	4 (13.3)
Systolic BP, median (IQR), mm Hg	140 (130–166.3)	147.5 (132.3–175.3)	139 (128–163.8)
Diastolic BP, median (IQR), mm Hg	86 (78.8–94)	88 (82–98.5)	82 (76.5–93)
Glucose, median (IQR), mmol/L	7.2 (6.2–7.8)	7.0 (5.8–7.8)	7.4 (6.5–7.8)
Procedural characteristics			
Anesthesia type, n (%)			
Local anesthesia with or without sedation	58 (96.7)	29 (96.7)	29 (96.7)
General anesthesia	2 (3.3)	1 (3.3)	1 (3.3)
Onset-to-recanalization time, median (IQR), min	465 (373.8–683.8)	415 (332.5–737.5)	482.5 (395–661.3)
Puncture to recanalization time, median (IQR), min	94 (67.3–110)	89.5 (63.5–108)	97.5 (75.8–112.3)
Onset-to-recanalization time, median (IQR), min	575 (448.3–786.5)	510 (421.3–839.3)	600 (462.8–767.5)
Recanalization-to-immediate MRI time, median (IQR), min*	74.5 (52.8–83.3)	74.0 (59.0–81.5)	75.0 (61.0–82.0)
No. of passes, n (%)			
≤2 passes	47 (78.3)	23 (76.7)	24 (80.0)
>2 passes	13 (21.7)	7 (23.3)	6 (20.0)

Data are presented as mean (SD), median (IQR), or n (%), as appropriate. AF indicates atrial fibrillation; ASPECTS, Alberta Stroke Program Early Computed Tomography Score; BP, blood pressure; CAD, coronary artery disease; CE, cardioembolism; ICA, internal carotid artery; IPostC, ischemic postconditioning; IQR, interquartile range; IV, intravenous; LAA, large-artery atherosclerosis; M1, M1 segment of the middle cerebral artery; MRI, magnetic resonance imaging; mRS, modified Rankin Scale; NIHSS, National Institutes of Health Stroke Scale; and TOAST, Trial of Org 10172 in Acute Stroke Treatment.

*Recanalization-to-immediate MRI time was assessed in the serial MRI subgroup (n=30; 15 per group).

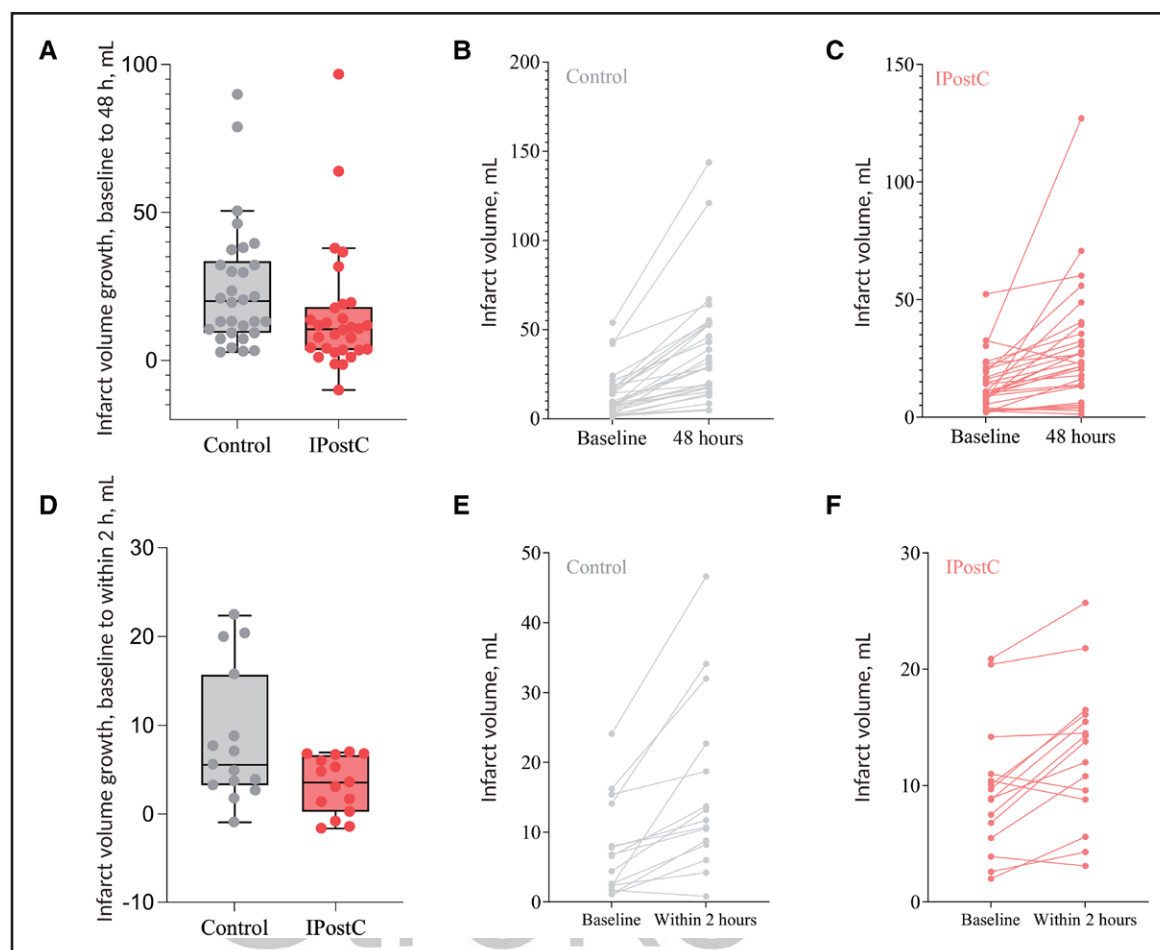


Figure 3. Infarct growth in the ischemic postconditioning (IPostC) and control groups.

A, Infarct growth from baseline to 48 hours in the control and IPostC groups. **B** and **C**, Paired patient-level changes in infarct volume from baseline to 48 hours in the control group (**B**) and IPostC group (**C**). **D**, Early infarct growth from baseline to within 2 hours after the procedure in the control and IPostC groups. **E** and **F**, Paired patient-level changes in infarct volume from baseline to within 2 hours after the procedure in the control group (**E**) and IPostC group (**F**).

Secondary Outcomes

Secondary imaging and clinical outcomes are summarized in Table 2.

In the serial MRI subgroup ($n=30$; 15 per group), early infarct growth was lower in the IPostC group than in the control group (median [IQR], 3.6 mL [0.9–6.4] versus 5.6 mL [3.5–12.3]; median difference, -3.2 mL [95% CI, -8.7 to -0.1]; $P=0.045$; Figures 3D through 3F). Using the same post hoc median regression model, the between-group difference remained in favor of IPostC (adjusted median difference, -5.3 mL [95% CI, -10.1 to -0.4]; $P=0.034$). Late infarct growth and immediate postprocedure infarct volume did not differ significantly between groups in the serial MRI subgroup. In the overall cohort, 48-hour infarct volume also did not differ significantly between groups.

Among clinical secondary outcomes, NIHSS scores at 24 hours were lower in the IPostC group than in the control group (median [IQR], 8.5 [6.3–11.0] versus 11.0 [9.3–15.5]; median difference, -3.0 [95% CI,

-5.0 to -1.0]; $P=0.005$; Figure 4). In a post hoc linear mixed-effects model adjusted for baseline NIHSS score, with a patient-level random intercept, the mean follow-up NIHSS score was 1.8 points lower in the IPostC group than in the control group (95% CI, -3.5 to -0.1 ; $P=0.040$). NIHSS scores at 72 hours and at 120 hours or discharge did not differ significantly between groups. The distribution of 90-day mRS scores is shown in Figure 5, with no significant shift between groups (common odds ratio, 1.1 [95% CI, 0.4–2.6]; $P=0.921$). Rates of good outcome, excellent outcome, and early neurological improvement did not differ significantly between groups.

Safety Outcomes

Safety outcomes are summarized in Table 3. The as-treated population was identical to the intention-to-treat population. No IPostC-related arterial dissection, vessel perforation, or distal embolization occurred. Vasospasm was observed in 1 of 30 patients in the IPostC group.

Table 2. Primary and Secondary Outcomes

Infarct volume outcomes	IPostC	Control	Effect size (95% CI)	P value
	n=30	n=30		
Baseline infarct volume, median (IQR), mL*	10.3 (5.8 to 18.7)	8.2 (4.7 to 16.4)	...	...
48-h infarct volume, median (IQR), mL	23.1 (13.5 to 34.7)	29.2 (17.4 to 52.8)	−5.8 (−16.8 to 4.2)†	0.252‡
Infarct growth from baseline to 48 h, median (IQR), mL	10.5 (3.6 to 16.8)	20.1 (9.7 to 32.3)	−8.7 (−17.1 to −1.4)†	0.015‡
Immediate postprocedure infarct volume, median (IQR), mL§	13.8 (9.2 to 15.8)	11.7 (8.5 to 20.7)	0.0 (−7.9 to 4.8)†	1.000‡
Early infarct growth, median (IQR), mL§	3.6 (0.9 to 6.4)	5.6 (3.5 to 12.3)	−3.2 (−8.7 to −0.1)†	0.045‡
Late infarct growth, median (IQR), mL§	9.6 (2.8 to 12.2)	7.3 (4.3 to 18.9)	2.1 (−5.3 to 9.1)†	0.412‡
Functional outcomes				
90-d mRS score, median (IQR)	2 (1 to 3)	2.5 (1 to 3)	1.1 (0.4 to 2.6)¶	0.921
Good outcome at 90 d (mRS score 0 to 2), n (%)	16 (53.3)	15 (50.0)	1.1 (0.7 to 1.7)¶¶	0.796
Excellent outcome at 90 d (mRS score 0–1), n (%)	10 (33.3)	10 (33.3)	1.0 (0.5 to 2.0)¶¶	1.000
Neurological outcomes				
NIHSS score at 24 h, median (IQR)	8.5 (6.3 to 11)	11 (9.3 to 15.5)	−3.0 (−5.0 to −1.0)†	0.005‡
NIHSS score at 72 h, mean (SD)	8.7±2.6	10.5±4.7	−1.7 (−3.7 to 0.2)‡¶	0.081
NIHSS score at 120 h or discharge, mean (SD)	7.6±3.1	8.1±3.7	−0.5 (−2.3 to 1.2)‡¶	0.546
Early neurological improvement, n (%)	4 (13.3)	2 (6.7)	2.0 (0.4 to 10.1)¶¶	0.671

Data are presented as mean (SD), median (IQR), or n (%), as appropriate. Effect sizes are unadjusted unless otherwise specified. IPostC indicates ischemic postconditioning; IQR, interquartile range; mRS, modified Rankin Scale; and NIHSS, National Institutes of Health Stroke Scale.

*Baseline infarct volume is shown to contextualize the primary outcome and was not an outcome measure.

†Hodges-Lehmann median difference.

‡Mann-Whitney *U* test.

§Analyses were performed in the serial MRI subgroup (n=30; 15 per group).

¶Common odds ratio from ordinal logistic regression.

¶¶Risk ratio.

#Mean difference.



Final expanded Thrombolysis in Cerebral Infarction grade after IPostC was $\geq 2b$ in all patients, and no rescue therapy was required after the IPostC procedure. Target-vessel recanalization status on 48-hour magnetic resonance angiography did not differ significantly between groups. Rates of all-cause mortality within 90 days, symptomatic intracranial hemorrhage, and any intracranial hemorrhage were also similar between groups.

Sensitivity Analysis

After excluding patients with target-vessel reocclusion on 48-hour magnetic resonance angiography (n=3), infarct growth from baseline to 48 hours remained lower in the IPostC group than in the control group (median difference, −8.3 mL [95% CI, −16.1 to −0.9]; $P=0.022$).

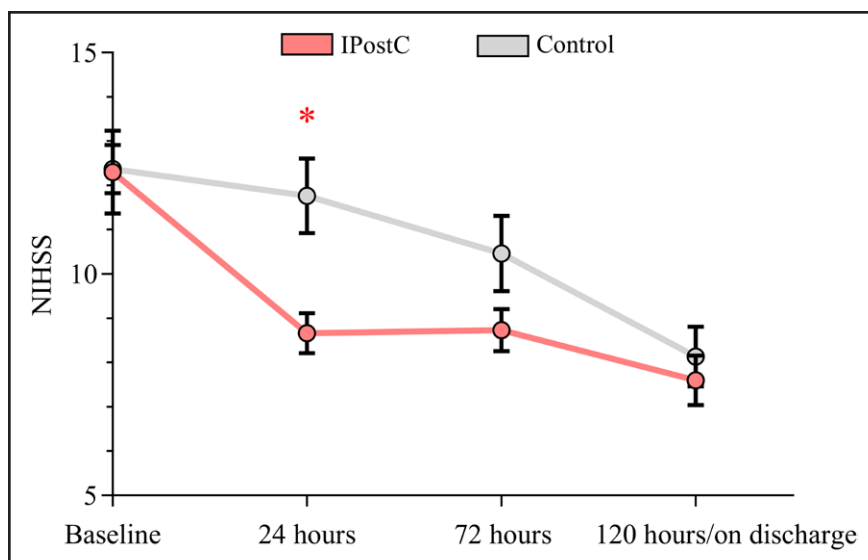


Figure 4. Changes in National Institutes of Health Stroke Scale (NIHSS) scores during hospitalization.

Changes in NIHSS scores from baseline to 120 hours or discharge in the ischemic postconditioning (IPostC) and control groups. Error bars indicate the SEM. The asterisk indicates a significant between-group difference at 24 hours.

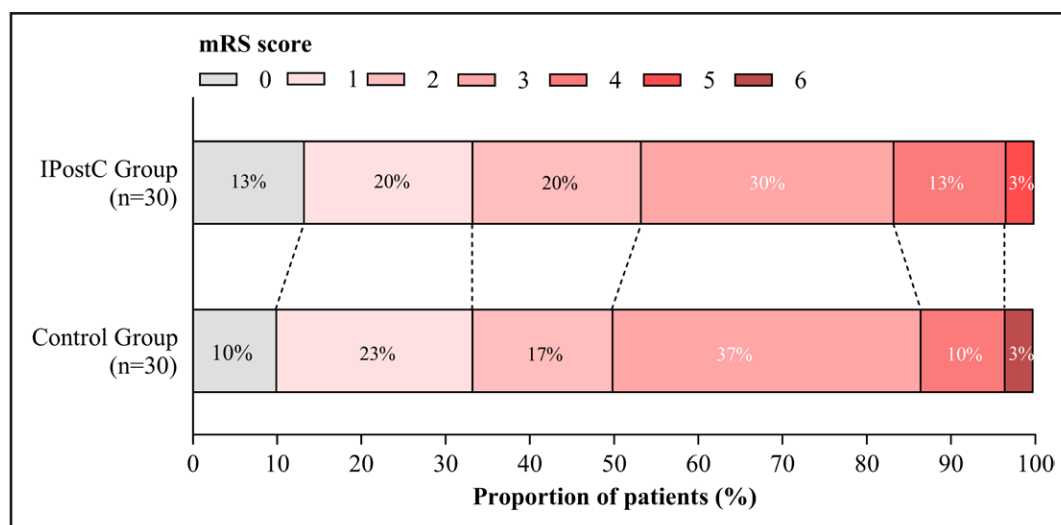


Figure 5. Distribution of modified Rankin Scale (mRS) scores at 90 days.

mRS scores range from 0 to 6, with higher scores indicating increasing disability, and 6 indicating death. IPostC indicates ischemic postconditioning.

Exploratory DCE-MRI Analysis

In the serial MRI subgroup, BBB permeability was assessed using DCE-MRI-derived Ktrans. Baseline characteristics are shown by DCE-MRI availability in Table S1 and by treatment assignment within the DCE-MRI subgroup in Table S2. Infarct-core Ktrans did not differ significantly between the IPostC and control groups (mean±SD, 1.0±0.6 versus 1.2±0.5; $P=0.297$). Peri-infarct Ktrans was lower in the IPostC group than in the control group (mean±SD, 1.2±0.5 versus 1.9±0.3; $P<0.001$).

DISCUSSION

This pilot, proof-of-concept randomized controlled trial suggests that IPostC, performed as 4 cycles of 2-minute balloon inflation followed by 2-minute deflation immediately after successful recanalization, reduced infarct growth from baseline to 48 hours, and was associated with lower NIHSS scores at 24 hours. However, 90-day mRS score outcomes did not differ between groups, possibly because early reductions in infarct growth and neurological deficits may not translate into long-term functional improvement, particularly in a trial of this size. In the serial MRI subgroup ($n=30$), early infarct growth was lower with IPostC. In addition, peri-infarct BBB permeability at 48 hours was lower in the IPostC group, consistent with prior experimental evidence suggesting preservation of BBB integrity.^{20–22}

Prior clinical studies have also suggested that IPostC may reduce infarct growth. Deng et al¹⁶ reported a 12.2 mL reduction in infarct growth with IPostC, broadly consistent with the 8.7 mL reduction observed in the present study. However, the estimate by Deng et al¹⁶ was based on baseline CTP and 24-hour noncontrast computed tomography, whereas the present study used MRI-based

infarct volume assessment. In our earlier MRI-based observational cohort, IPostC was associated with reduced infarct growth in the deep gray matter, although total infarct growth in the overall ischemic region did not differ significantly.¹⁵ Deep gray matter structures are supplied by small perforating end arteries with limited collateral support,^{23,24} making them vulnerable to microvascular no-reflow after recanalization. Whether the reduced infarct growth observed with IPostC is mediated by improved microvascular reperfusion remains to be determined.

Early infarct growth was lower in the IPostC group in the serial MRI subgroup. Prior serial MRI data have shown that, among patients with successful recanalization, early infarct growth is influenced by baseline DWI lesion volume and the number of thrombectomy passes.⁵ In the present trial, these variables were well balanced between groups, and the reduction in early infarct growth remained significant in the adjusted analysis. These findings support a potential early cytoprotective effect of IPostC. Because early infarct growth may result from tissue injury occurring both before and after recanalization, and because IPostC was performed only after recanalization, the observed reduction is more plausibly attributable to attenuation of ischemia-reperfusion injury. Preclinical studies suggest that IPostC may reduce ischemia-reperfusion injury through attenuation of oxidative stress, inflammation, apoptosis, BBB disruption, and microvascular injury.^{11,12} However, the precise in-human mechanism remains uncertain and warrants further investigation.

The optimal IPostC regimen remains uncertain. Our previous 3+3 dose-escalation study showed that in situ IPostC, performed as 4 balloon inflation-deflation cycles with a duration of up to 5 minutes, was feasible, safe, and well tolerated.¹⁴ However, that study was designed primarily to assess safety and tolerability rather than to optimize the treatment regimen. A recent adaptive dose-finding

Table 3. Safety Outcomes

	Total	IPostC	Control
No. of patients	60	30	30
IPostC-related procedural complications, n (%)			
Arterial dissection	...	0 (0.0)	...
Vessel perforation	...	0 (0.0)	...
Distal embolization	...	0 (0.0)	...
Vasospasm	...	1 (3.3)	...
Final postprocedure eTICI grade, n (%)			
2b	13 (21.7)	6 (20.0)	7 (23.3)
2c	5 (8.3)	2 (6.7)	3 (10.0)
3	42 (70.0)	22 (73.3)	20 (66.7)
Target-vessel recanalization at 48 h, n (%)	57 (95.0)	30 (100.0)	27 (90.0)
All-cause mortality within 90 d, n (%)	1 (1.7)	0 (0.0)	1 (3.3)
slCH, n (%)	2 (3.3)	0 (0.0)	2 (6.7)
Any intracranial hemorrhage, n (%)	17 (28.3)	7 (23.3)	10 (33.3)

Data are presented as n (%). Ellipsis indicates not applicable. The as-treated population was identical to the intention-to-treat population. IPostC-related procedural complications were assessed only in the IPostC group. Final postprocedure eTICI grade refers to the final angiographic grade after the assigned procedure. Target-vessel recanalization status was assessed on 48-hour MRA. eTICI indicates expanded Thrombolysis in Cerebral Infarction; IPostC, ischemic postconditioning; MRA, magnetic resonance angiography; and slCH, symptomatic intracranial hemorrhage.

study suggested that rapid local IPostC using 4 cycles of 2-minute balloon inflation followed by 2-minute balloon deflation may provide a favorable efficacy-toxicity profile.²⁵ Nevertheless, because that study used ICA conditioning rather than in situ conditioning, the optimal dosing strategy for in situ IPostC remains undefined. Moreover, most available IPostC regimens have empirically used 4 or 5 cycles, and no dedicated dose-finding study has evaluated the optimal number of cycles.^{14–16,25} Future studies should therefore assess both cycle duration and cycle number to define the optimal IPostC regimen.

The optimal conditioning site for IPostC also requires further study. ICA-based rapid local IPostC is technically simpler and potentially safer because it is performed in a larger proximal artery.¹⁶ However, collateral pathways through the circle of Willis, particularly the anterior and posterior communicating arteries, may maintain distal perfusion during ICA balloon inflation, resulting in incomplete and patient-dependent flow interruption within the target vascular territory. Conversely, in patients with poor primary collaterals, temporary ICA occlusion may expose broader nontarget anterior circulation territories to additional ischemia. In situ IPostC may more directly modulate reperfusion at the target arterial segment, but it raises greater concern for vessel-wall injury. This risk may be mitigated by low-pressure balloon inflation and careful balloon sizing, consistent with our previous large-animal study showing no significant vascular wall injury after repeated balloon inflation-deflation cycles.¹⁴ Future

studies should compare in situ IPostC with ICA-based rapid local IPostC to define the optimal conditioning site.

This study has several limitations. First, as a single-center pilot trial with 30 patients per group, it was not powered to detect modest differences in functional outcomes or to precisely evaluate infrequent safety events. Second, the fixed IPostC regimen precluded assessment of dose-response effects related to inflation-deflation duration, cycle number, or conditioning site. Third, the proportion of women was relatively low (18%), which may limit generalizability of the findings. Finally, no adjustment was made for multiple comparisons; therefore, the findings should be interpreted as hypothesis-generating.

In conclusion, this proof-of-concept trial met its primary outcome, showing reduced infarct growth with IPostC, whereas 90-day functional outcomes did not differ significantly between groups in this pilot sample. These findings support further evaluation in larger, adequately powered trials. The current pilot data, together with the phase 2b PROTECT-2 trial, will inform the design of a definitive phase 3 trial.

ARTICLE INFORMATION

Received March 28, 2026; final revision received June 16, 2026; accepted June 29, 2026.

Presented in part at the International Stroke Conference, New Orleans, LA, February 4–6, 2026.

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Acknowledgments

The authors thank all patients and their representatives for their participation in this trial.

Sources of Funding

This study was supported by the National Health Commission Capacity Building and Continuing Education Center Nervous System and Minimally Invasive Intervention Program (No. GWJJ2022100106); the Tianjin Health Science and Technology Project (Nos. MS20015 and TJWJ2022QN063); the National Natural Science Foundation of China (No. 82571507); and the Key Science and Technology Project for Stroke Prevention and Treatment (No. 2025PSPT0705100).

Disclosures

Dr Fisher reports compensation from Lumosa for consultant services; compensation from Takeda Pharmaceutical Company for data and safety monitoring services; compensation from Simcere USA for consultant services; and compensation from Revalio for consultant services. The other authors report no conflicts.

Supplemental Material

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